WEEK 7 · LECTURE L3 · UNIT III

Auditing an Algorithm: What a Fairness Audit Involves

How to examine an HR algorithm before trusting it: scope, evidence, error rates, the four fifths test, documentation and governance.

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AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

1. Explain why accuracy hides what matters, and read a confusion matrix by group.
2. Compute and interpret the four fifths ratio, and distinguish selection rate checks from error rate checks.
3. Design an audit in steps mapped to the NIST AI RMF, using SMACTR, model cards and datasheets.
4. Evaluate the limits of audits and documentation as safeguards.

ONE SOURCE

The same theories as the in class lecture and the course hub, without the simulation data used in class. Course outcome: CLO 3.

COMMIT BEFORE THEORY

Commit to an answer first

A vendor says its resignation model is 92% accurate. In the firm, 8% of staff resign each year. What does 92% accuracy tell you?

A

That the model is excellent

B

Almost nothing: a model that predicts nobody will resign is also 92% accurate

C

That 92% of resignations will be caught

D

That the model is fair

EXAM LENS

The answer is B. When outcomes are rare, accuracy rewards predicting the common outcome for everyone.

ACCURACY HIDES WHAT MATTERS

The confusion matrix

	Actually left	Actually stayed
Predicted to leave	True positive	False positive: wrongly flagged
Predicted to stay	False negative: missed leaver	True negative

Compare across groups

Recall: share of leavers caught. Precision: share of flags that are right. False positive rate: who is wrongly flagged. False negative rate: who is missed.

TWO KINDS OF FAIRNESS CHECK

Selection rates and error rates

Selection rates

Does the tool select one group at a lower rate? The four fifths rule: an impact ratio below 0.80 flags adverse impact.

Error rates

Does the tool make more mistakes for one group: more false flags, or more misses? Compare the rates group by group.

EXAM LENS

A tool can pass one check and fail the other; Week 9 shows why both cannot always be equalized at once.

An illustrative calculation

	Female	Male
Applications	1,000	1,000
Offers	80	140
Selection rate (offers ÷ applications)	8.0%	14.0%
Impact ratio (lower ÷ higher rate)	8.0 ÷ 14.0 = 0.57	

EXAM LENS

A ratio below 0.80 signals adverse impact. Compare selection rates, never raw offer counts, and remember the rule is a rule of thumb: small samples need statistical tests too.

Illustrative figures (Course framing).

THE AUDIT

Eight steps, tagged by NIST function

Step	NIST function
1 · Define the decision and the groups affected	Map
2 · Map every stage where the tool acts	Map
3 · Compute selection rates by group at each stage	Measure
4 · Apply the four fifths test	Measure
5 · Check error rates and proxies among inputs; check data protection duties	Measure
6 · Decide: keep, fix, add human review, or stop	Manage
7 · Set human review and appeal	Manage
8 · Assign ownership and re audit dates	Govern

NIST (2023)

The AI Risk Management Framework

Govern

Culture, policies, accountability; runs through the other three.

Map

Context, and who may be harmed.

Measure

Analyze and track risks.

Manage

Prioritize and act on risks.

EXAM LENS

Trustworthy AI characteristics: valid and reliable; safe; secure and resilient; accountable and transparent; explainable and interpretable; privacy enhanced; fair with harmful bias managed. An audit is Measure: without Govern (a named owner) and Manage (a decision) it changes nothing. Voluntary guidance: it says what to do, not how much is enough.

RAJI ET AL. (2020)

SMACTR: an internal audit in five stages

Scoping

Purpose, use case, principles and risks of harm.

Mapping

Stakeholders, collaborators and who is affected.

Artifact collection

Documents such as model cards and datasheets.

Testing

Tests of performance and harm, including by group.

Reflection

Risks, mitigations, remediation and sign off.

EXAM LENS

Audit across the whole development life cycle, not only after launch. Internal auditors may lack independence from the teams they review: contrast with external or regulatory audit.

DOCUMENTATION

Model cards and datasheets

Model cards (Mitchell et al., 2019)

Intended use and uses it was not designed for; factors across which performance may differ; metrics and why; evaluation data; results disaggregated by group and intersection; ethical considerations and caveats.

Datasheets (Gebru et al., 2021)

Why the data were collected, how, by whom, who is represented, and recommended uses. If a hiring model learned from mostly male past hires, a datasheet would reveal it: Barocas and Selbst’s labeled examples route.

EXAM LENS

A model without a model card cannot be audited properly. But a card is only as honest as its authors, it does not test the model, and documentation rarely happens without a requirement: documentation enables accountability; it does not ensure it.

APPLY

Auditing a vendor’s screening tool

Teaching case (Course framing)

A Chennai insurer plans to use a vendor tool that scores 40,000 applicants a year for sales roles. The vendor reports 88% accuracy and offers no breakdown by group. About 30% of applicants are women and many apply in Tamil. The vendor says the model is proprietary.

Work it through

1. Why is 88% accuracy not enough?
2. Which two fairness checks would you run first?
3. Walk through SMACTR: what would you scope and map?
4. Which model card items would you insist on?

COMMON CONFUSIONS

Tempting readings, and better ones

A tempting reading	A better reading
High accuracy means a good and fair model	Accuracy can hide rare outcomes and unequal errors; compare rates by group
An audit happens after launch	SMACTR audits across the life cycle
A proprietary model cannot be audited	Demand model cards, disaggregated results and testing access before buying
Compare offer counts by group	Compare selection rates, each on its own group’s applications

COMPARE AND EVALUATE

Five tools for auditing an algorithm

Tool	Use it to	Limit
Confusion matrix	Compare error rates by group	Needs known outcomes
Four fifths rule (1978)	Flag adverse impact in selection	Rule of thumb; small samples mislead
NIST AI RMF (2023)	Organize risk work and ownership	Voluntary; no thresholds
SMACTR (2020)	Run an internal life cycle audit	Independence
Model cards; datasheets	Demand documentation	Only as honest as the authors

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse

An Indian bank’s HR team proposes an algorithm to shortlist candidates for promotion. Using the confusion matrix, the four fifths rule, SMACTR and the NIST AI RMF, design an audit before the tool is used, and state one limit of your audit. **20 marks**

A strong answer: explains why accuracy is insufficient and which error rates to compare; applies the four fifths rule correctly; sequences the audit through SMACTR with artifacts; uses Govern to assign ownership; notes a limit such as internal bias.

“Documentation is not accountability.” Evaluate model cards, datasheets and internal audits as safeguards. **20 marks**

A strong answer: explains each; notes honesty and independence limits; argues they enable but do not ensure accountability.

Calculate an impact ratio from given selection data and interpret it. **10 marks**

A strong answer: uses rates, not counts; compares with 0.80; notes sample size caution.

CARRY THESE INTO THE EXAMINATION

Four things to remember

1. **Accuracy hides rare outcomes.**
   Read the confusion matrix, by group.

2. **Two checks: selection rates and error rates.**
   Rates, never raw counts.

3. **An audit is Measure; it needs Govern and Manage.**
   A named owner and a decision.

4. **Demand documentation before you trust a model.**
   And remember it does not test the model.

SET READINGS · FROM THE LESSON PLAN

Read for this lecture

- EEOC Uniform Guidelines on Employee Selection Procedures (29 CFR 1607).
- NIST (2023). AI Risk Management Framework 1.0.

NEXT WEEK

- Johnson, Carlson and Kavanagh (2025), Chapter 11; Angrave et al. (2016); Knaflic (2015).

> **NOTE**
> The other works named in this lecture are theory sources for reference, not set readings.

Sources for this lecture

- Equal Employment Opportunity Commission and others. (1978). *Uniform guidelines on employee selection procedures*. 29 C.F.R. part 1607.
- Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., Wallach, H., Daumé III, H., & Crawford, K. (2021). Datasheets for datasets. *Communications of the ACM, 64*(12), 86 to 92. https://doi.org/10.1145/3458723
- Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., Spitzer, E., Raji, I. D., & Gebru, T. (2019). Model cards for model reporting. *Proceedings of the Conference on Fairness, Accountability, and Transparency*, 220 to 229. https://doi.org/10.1145/3287560.3287596
- National Institute of Standards and Technology. (2023). *Artificial intelligence risk management framework (AI RMF 1.0)* (NIST AI 100-1).
- Raji, I. D., Smart, A., White, R. N., Mitchell, M., Gebru, T., Hutchinson, B., Smith-Loud, J., Theron, D., & Barnes, P. (2020). Closing the AI accountability gap. *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, 33 to 44. https://doi.org/10.1145/3351095.3372873